

NLP Project Report: Routing of emails/tickets to different departments using LLMs

Anna Smetanina

The goal of the project is to automatically route the entering emails from the clients into one of the 5 company departments: Technical support, Customer service, Billing and payments, Sales and pre-sales and General inquiry. The task is a multi-class classification problem. The preprocessed input is using the email subject + the email body, and the target is the email department (queue).

We compare generative and discriminative approaches:

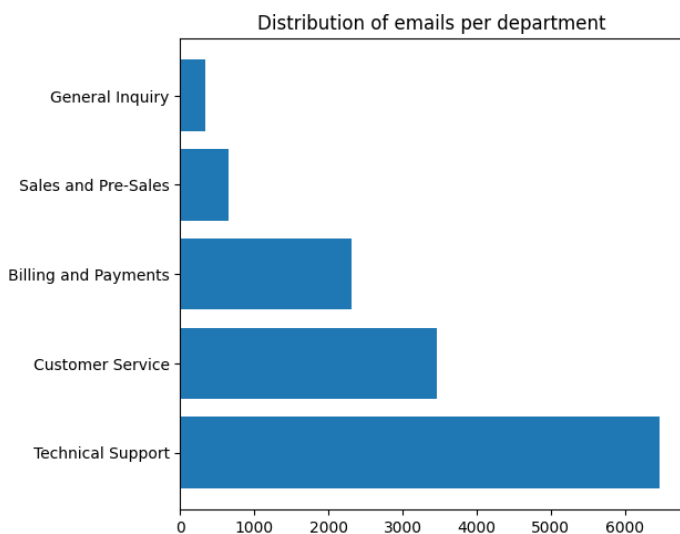
1. generative classification with frozen GPT2 and DistillGPT2 models,
2. generative classification with GPT2 and DistillGPT2 models after parameter-efficient fine-tuning with LoRA,
3. discriminative classification with DistillBERT
4. machine learning method TD-IDF with Logistic regression

All the methods are evaluated on the same test set. It is interesting to compare these approaches because the models are different by the complexity and by the type of the architecture (encoder and decoder). We compare if generative decoder is better than discriminative encoder for the routing text.

Dataset

The dataset used is Tobi-Bueck/customer-support-tickets (Hugging Face Datasets). Preprocessing is done by the `load_and_prepare_data` in the file `datapreparation.py`.

Train set size has 13249, validation set 1656 and test set 1657 emails. As can be seen from the bar chart, the dataset is imbalanced. The most frequent class in the training set is “Technical support” (6476 emails) and the least frequent class is “General inquiry” (340 emails). So if a model learns to predict every label as the most frequent class the baseline accuracy would be about 0.51.



Methods

Padding in GPT2, DistillGPT2 with LoRA versions is done by using the EOS token, because GPT2 does not have padding natively. Shorter sequences are padded from the left side, to the maximum

allowed length and attention mask is created by 1 for real tokens and 0 for padding tokens so the useful tokens do not attend to the empty ones.

For the DistillBERT padding is handled automatically by the Hugging Face tokenizer.

The best model during training is chosen on the validation set, the models weights are saved in the epoch if it shows highest validation accuracy. Then the best models are inferenced on the test set.

Method 1: frozen Gpt2+prompting

This is the most simple method that does not require any training. The used model is GPT-2 with pretrained weights. It gets an instruction prompt with an instruction, the names of the departments, subject and email body: “Route the following customer email to exactly one department. Valid departments:

Technical Support
Customer Service
Billing and Payments
Sales and Pre-Sales
General Inquiry

Subject: {subject}

Body: {body}

Department:”.

GPT-2 is an autoregressive decoder model so it generates the continuation of the text as the most probable token given the context and then the answer is converted into one of 5 labels.

Method 2: frozen DistillGpt2+prompting

DistillGPT2 is a smaller distilled version of GPT2, which has fewer parameters and uses less compute. This experiment checks if a smaller model shows a trade-off: whether it is worse in accuracy and faster in inference.

Method 3: GPT-2 + LoRA

The same GPT-2 model fine-tuned using the LoRA (Low-Rank Adaptation) method. The weights of the full model are frozen and we learn the small residual matrices that are added to the attention projection. Only a small number of parameters is trained so it is a cost and time-efficient method.

LoRA configuration: rank = 8, lora_alpha = 16, lora_dropout = 0.05, lr = 2e-4, max_epochs = 8, early_stopping_patience = 2, batch_size = 32.

Only 294,912 parameters are learned, so with respect to the full gpt2 it is about 0.2% of all weights, following the idea of PEFT (parameter-efficient tuning).

Loss

Each training example is prompt + the correct queue. The loss is calculated as a Cross-Entropy loss.

Method 4: DistillGPT-2 + LoRA

The same LoRA configuration but on DistillGPT2, the same targets and the same loss. Here we can compare how the size of the base model influences when fine-tuning is used.

Method 5: DistillBERT classifier

DistillBERT is an encoder model. The input of the model is the same, subject + body. For the output generation it uses a classification head. The outputs are the logits, raw output scores for each class which are then passed into the softmax function. The final prediction is a class with a maximum

probability. This is a classical discriminative method for text classification, the model learns to discriminate the classes, not to predict the next text. Usually it produces the best result on such tasks.

The configuration for DistilBERT:

`lr = 2e-5, max_epochs = 8, early_stopping_patience = 2, batch_size = 64`

Method 6: TF-IDF and Logistic Regression

Each email is represented with TF-IDF features for unigram and bigrams, after the logistic regression classification is applied. This is a classical ML approach which allows us to see if the use of Transformers is giving a meaningful gain in evaluation metrics for our task with respect to this classical baseline.

Results

All the methods are compared on the same test set of 1657 emails. To compare the results, we used accuracy, training time, inference time and peak CUDA allocation.

	accuracy	training_time_s	inference_time_s	training_peak_cuda_MB	inference_peak_cuda_MB
GPT2 prompting	0.242004	0.000000	6.443411	NaN	1609.177600
DistilGPT2 prompting	0.424261	0.000000	3.245651	NaN	1163.062784
GPT2 + LoRA	0.674110	930.379049	5.822641	22637.511680	1443.113984
DistilGPT2 + LoRA	0.652987	515.858416	3.342007	15643.642880	1083.378176
DistilBERT classifier	0.836451	417.611172	2.521496	9858.120192	1726.191616
TF-IDF + LogReg	0.751358	16.784804	0.228255	NaN	NaN

Main result is achieved by DistilBERT classifier with an accuracy of 0.836. The second best is TF-IDF with a logistic regression. LoRA fine-tuning made the accuracy between GPT2 and DistilGPT2 more similar between each other (0.674 vs 0.653) especially compared to their baseline frozen model versions (0.242 and 0.424). Both basic models showed worse results than the majority baseline class (Technical Support) with its 0.512. So it means that using a GPT2 with instruction prompt for the task of text routing is not suitable.

TF-IDF with logistic regression performed better than both LoRA models. It had also the fastest training time of 16.78 seconds and the fastest inference time of 0.22 seconds. Together with its accuracy of 0.751 it looks like a reasonably good choice for production in real-world scenarios with routing emails if the resources are limited. Being a classical ML method, it also was trained and inferenced on the CPU, not on the GPU as the other models.

The macro-F1 score, precision, recall and confusion matrices are included in the notebook as additional evaluation diagnostics, since the classes are imbalanced in the dataset. From there we can see that accuracy of the LoRA models is almost zero in the underrepresented classes "General Inquiry" and "Sales and presales".

Conclusions

This project compared generative, discriminative and classical machine learning approaches for the email routing. The best result was shown by a discriminative classifier DistilBERT which confirms that it is the best method for multi-class classification. LoRA improved the frozen models results but did not reach a better result than classical machine learning linear classifier.

The main conclusion is that choosing the right model depends on the task formulation and the available resources. Bigger architecture model does not mean that the task will be guaranteed better accuracy, other methods should also be investigated.

Future works can include using methods for class balancing, e.g. oversampling or class importance, prompt engineering or testing other model architectures, especially from the BERT family since it has proven to be the most accurate model, or experimenting with other linear ML models.